

KARUPAKALA RAHUL CHANDRA

Hyderabad, India | +91 9502568861 | rahulchandrakarupakala@gmail.com
linkedin.com/in/rahul-chandra-karupakala-318a5b326 | github.com/k-rahul-chandra

EDUCATION

Malla Reddy University, Hyderabad, India
B.Tech in Artificial Intelligence & Machine Learning (AIML) (*Expected 2027*)
CGPA: 7.5 / 10

TECHNICAL SKILLS

Languages:	Python (Intermediate to Advanced), Java, JavaScript, SQL
AI / Machine Learning:	Scikit-Learn, Unsupervised Clustering, Predictive Modeling, Pandas, NumPy
GenAI & Agents:	LangChain, LLMs (OpenAI API / Ollama), CrewAI / AutoGen, Prompt Engineering
Databases & Vectors:	MySQL, ChromaDB / Pinecone (Vector Databases)
Tools & Web Dev:	Git, GitHub, Streamlit, Node.js, React.js, HTML5, CSS3

TECHNICAL PROJECTS

J.A.R.V.I.S. – Context-Aware Voice AI Agent (Solo Project)

Python, LangChain, OpenAI/Ollama, Whisper API, PyAutoGUI

- Designing and engineering an autonomous local AI assistant utilizing Large Language Models (LLMs) and Advanced Function Calling to manage system-level operations.
- Implementing a real-time Speech-to-Text and Text-to-Speech pipeline using Whisper API for ultra-low latency voice interaction.
- Integrating automation scripts to control system applications, manage background calendars, and dynamically trigger text dispatches via messaging APIs based on raw voice intent.

Enterprise Bank Customer Segmentation System (Academic Collaboration)

Python, Scikit-Learn, Pandas, NumPy, Streamlit, Git

- Developed an end-to-end unsupervised machine learning pipeline to extract actionable consumer profiles from a massive financial dataset containing 1M+ transactions across 800K Indian consumers.
- Applied RFM (Recency, Frequency, Monetary) scaling and optimized K-Means clustering using the Elbow Method to target highly accurate demographic clusters.
- Handled comprehensive data preprocessing, missing value imputation, and feature scaling to overcome hardware memory constraints during large-scale computational execution.
- Built and deployed an interactive executive visualization dashboard via Streamlit to visually isolate cluster properties and customer behaviors.

Multi-Agent Collaborative AI Research Assistant (Solo Project)

Python, CrewAI, LangChain, ChromaDB, Hugging Face

- Engineered a local multi-agent network where independent AI personas collaborate autonomously to conduct deep academic research and complex code debugging tasks.
- Programmed a multi-layered workflow consisting of a Research Agent to parse documentation, a Developer Agent to write optimized code, and a Reviewer Agent to stress-test scripts.
- Integrated ChromaDB as an embedded local vector database to empower agents with native semantic search capabilities over dynamic local repositories.

LEADERSHIP & COLLABORATION

Project Team Lead – Malla Reddy University

Academic Project Delivery & Version Control

- Directed a 3-member peer team during iterative software build cycles, establishing precise project timelines and delegating tasks mapping directly to individual core developer skill sets.
- Facilitated clear version control habits by establishing structured GitHub rules, minimizing deployment code conflicts, and overseeing smooth merge pipelines.